

VATTENTION: VERIFIED SPARSE ATTENTION

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ABSTRACT

State-of-the-art sparse attention methods for reducing decoding latency fall into two main categories: approximate top- k (and its extension, top- p) and recently introduced sampling-based estimation. However, these approaches are fundamentally limited in their ability to approximate full attention: they fail to provide consistent approximations across heads and query vectors and, most critically, lack guarantees on approximation quality, limiting their practical deployment. We observe that top- k and random sampling are complementary: top- k performs well when attention scores are dominated by a few tokens, whereas random sampling provides better estimates when attention scores are relatively uniform. Building on this insight and leveraging the statistical guarantees of sampling, we introduce vAttention, the first practical sparse attention mechanism with user-specified (ϵ, δ) guarantees on approximation accuracy (thus, “verified”). These guarantees make vAttention a compelling step toward practical, reliable deployment of sparse attention at scale. By unifying top- k and sampling, vAttention outperforms both individually, delivering a superior quality–efficiency trade-off. Our experiments show that vAttention significantly improves the quality of sparse attention (e.g., ~ 4.5 percentage points for Llama-3.1-8B-Inst and Deepseek-R1-Distill-Llama-8B on RULER-HARD), and effectively bridges the gap between full and sparse attention (e.g., across datasets, it matches full model quality with upto 20x sparsity). We also demonstrate that it can be deployed in reasoning scenarios to achieve fast decoding without compromising model quality (e.g., vAttention achieves full model quality on AIME2024 at 10x sparsity with up to 32K token generations). Code is open-sourced at <https://github.com/xAlgo-ai/sparse-attention-hub>.

1 INTRODUCTION

As the application of AI expands and workflows grow more complex, the volume of context that large language models (LLMs) must maintain is rapidly increasing (Touvron et al., 2023; Achiam et al., 2023; Liu et al., 2024a). However, Scaled Dot Product Attention (SDPA) operator, the core operation behind the success of transformer architectures (Vaswani, 2017; Brown et al., 2020), is not well suited for handling such long contexts during generation. Large contexts produce massive key–value (KV) embedding caches, and in autoregressive models, these caches must be repeatedly read for every new token prediction. This makes the decoding step inherently memory-bound and time-consuming (Kim et al., 2023). The problem becomes even more severe when the KV caches exceed available GPU memory and must be offloaded to CPU RAM, requiring costly transfers across the CPU–GPU boundary. These bottlenecks highlight a fundamental scalability issue in attention mechanisms, limiting the ability of LLMs to efficiently consider long contexts. A mitigation strategy is sparse attention, which reduces memory movement by attending only to a subset of tokens in the KV cache. A good sparse attention would offer highly accurate approximations of full attention that it replaces.

The core abstract problem in approximating Scaled Dot Product Attention (SDPA; see Eq. 3) is estimating the sum of n quantities (scalars for denominator and vectors for numerator in SDPA). Since, in hindsight, the tokens that contribute most to the attention output are those with the highest $a_i || \mathbf{v}_i ||_2$ (Desai et al., 2025), a natural approach is to choose tokens i with the largest query–key

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